Overview **Question** Test Cases Code

Instructions for Extended Match Questions
Use this slide to understand how to answer an EMQ

Instructions:
1. For each question, enter all the options which you find is the answer to that question as "**comma separated values**".
2. Enter the options in **ascending order** of option number.
3. An option can also be an answer to more than one question, or may not be an answer to any of the questions.
4. Please click here to view a sample template file for answering.
5. Please note that if you give an incorrect answer there will be a negative marking for the incorrect option.

List of Options:
1. T is is one to one.
2. T is onto.
3. T is an isomorphism.
4. $Nullspace(T) = \{(t, t, t) \mid t \in R\}$
5. $Nullspace(T) = \{(t, t, s) \mid t, s \in R\}$
6. $Nullspace(T) = \{(0, 0, 0)\}$
7. A basis of $Nullspace(T) = \{(1, 1, 1)\}$
8. A basis of $Nullspace(T) = \{(1, 1, 0), (0, 0, 1)\}$
9. $nullity(T) = 1$
10. $nullity(T) = 2$

List of Questions:

Q1 Consider the linear transformation $T: R^3 \to R^3$ defined by $T(x, y, z) = (x - y, y - z, 0)$. Which of the above options are true for T?

Q2 Consider the linear transformation $T: R^3 \to R^2$ defined by $T(x, y, z) = (x - y, y - z)$. Which of the above options are true for T?

Q3 Consider the linear transformation $T: R^3 \to R^3$ defined by $T(x, y, z) = (x - y, y - z, z - x)$.Which of the above options are true for T?

Q4 Consider the linear transformation $T: R^3 \to R^3$ defined by $T(x, y, z) = (x - y, 0, 0)$.Which of the above options are true for T?

Q5 Consider the linear transformation $T: R^3 \to R^3$ defined by $T(x, y, z) = (x, x + y, x + y + z)$.Which of the above options are true for T?